Instructor Checklist for Each Lab Practical Session

Course Section _____

Lab Practical Date _____

Version	Station	Quantity / Measurement to Record	Value
A (green)	1	Mass of solution (to 4 dec places)	20.3915 g
	4	Temperature to 0.1°C	22.3 °C
	7	Volume on buret to 0.01 mL	15.60 mL
B (yellow)	3	Eudiometer reading to 0.01 mL	31.40 mL
	9	i) Graduated cylinder volume to correct precision ii) Temperature to 0.1°C	31.50 mL 22.1° °C
C (purple)	3	i) Temperature to 0.1°C ii) Barometric reading as mmHg	20.8 °C 744.7 mmHg
	7	Buret to 0.01 mL	17.60 mL

CHEM 3A – Lab Practical – VERSION A

Name: Yusef AlZamZami

Section #: 19320

Student ID #: 1052687

*During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version A**.*

Station #1

11.1523 g

21.5293 g

$D = \frac{\text{Mass}}{\text{Volume}}$

$0.5180 = \frac{11.1523}{21.5293}$

0.5180

Station #2

Percent yield % = (actual yield)/ Therotical yield) x100

Alum= 8.879 g (7.4415/ 8.879) x 100 = 83.81 %

st. 7.4415 g

83.81 %

Station #3

$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$

Caculate the mass of hydrogen gas in grams if 0.0475 g Piece of magnesium?

0.5225 g

Station #4

It is 22°C Degress Celsius

22°C

Station #5

h = 1.2

w = 3.7

l = 3.8

V = l x w x h

in Cm^3

16.87 = 3.8 x 3.7 x 1.2

16.87

Station #6

Observations: It is gray cloudy color.

Reaction Type: This is a Redox Reaction

Balanced Equation: $CaCl_2 \times Na_2CO_3$

Station #7

Volume of G down Soda 10.00 mL NaOH concentration 0.9573 M

Initial Buret 1.27 mL

Final Buret = 8.73

8.73 mL

Station #8

1. Bunsen Burner
2. Graduated Cylinder

Station #9

The final Molarity by diluting 25.00 mL of a 2.5789 M sodium chloride soultion to a final volume of 100.00 mL ?

89.6844 mL

Station #10

21.9751 - 22.1715 = 0.1964 1st weighing

21.9751 - 22.1722 = 0.1971 2nd wieghing

1 = 0.1964 g 2 = 0.1971 g

Extra Space

CHEM 3A – Lab Practical – VERSION A **Name:** Ian Moua

Section #: _____ **Student ID #:** 1056547

*During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version A**.*

Station #1

empty = 14.4582

+ 24.8474

$\frac{24.8474}{100}$ = 24.8474

Station #2

$\frac{24.1233}{8.879} \cdot 100 = 271.6$

Station #3

$\frac{10.658 + 14.903}{0.0479} = 538.126\ g$

Station #4

24.0°C

Station #5

$3.6/16 \cdot 1\ 1/16 \cdot 1\ 1/16 = 1.0045 \approx 1.005\ cm^3$

Station #6

Observations: bubbly, white,

Reaction Type: acid/base

Balanced Equation: $Na_2CO_3 + CaCl_2 \rightarrow$ (Na2 + 2 CO4 Ca2)

Station #7

final $17\ 6/16$

$$\frac{0.9573 + 17\ 6/16}{1.27} = 5.9769$$

Station #8

1) Ignighting switch (101c)

2) Burret

Station #9

$$\frac{100.2589}{25.00} = 10.3156$$

Station #10

$$\frac{21.8563}{2} = 10.92815 \qquad \frac{21.8563 - 22.1715}{15.9999}$$

$$\frac{21.9751}{3} = 7.32503 \qquad = .0197$$

Extra Space

CHEM 3A – Lab Practical – VERSION A **Name:** VILLANUEVA, VICTOR

Section #: 19320 **Student ID #:** 1075713

*During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version A**.*

Station #1

10.0 mL

DENSITY = MASS / VOLUME

14.5904 g

20.7363 g

1.4212 g ANSWER

Station #2

VIAL A = 8.879 / 7.4415 g × 100 = 119.317 g

ANSWER

Station #3

0.0475 g = mg

$Mg(s) + 2\,HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$

0.095 g ANSWER ?

Station #4

21° CELCIUS

Station #5

V = l × w × h

3.7 cm L 1.2 cm W 1.2 cm H

5.328 cm^3

Station #6

Observations: CLOUDY WHITE SOLUTION WHEN MIXED

Reaction Type: ACID/BASE NEUTRALIZATION

Balanced Equation: $CaCl_2 + Na_2CO_3$

Station #7

BURET READING = 17.4 NaOH

17.4mL

NaOH CONTENT = 0.9573M

V = 18.17612034

Station #8

BUNSEN BURNER

VOLUMETRIC FLASK

Station #9

FINAL MOLARITY DILUTING 25.0mL

0.6447 mol

$\frac{2.5789\text{m} \quad 500\text{ml/m}}{25.00\text{mL}} = 0.6447\text{M}$

Station #10

MASS of EMPTY CRUCIBLE + LID 21.9563

ANSWER = 0.3159g

0.0079

mg 21.9751g

mg of compound 22.1715g

22.1722

Extra Space

CHEM 3A – Lab Practical – VERSION A **Name:** Marcos Argupta

Section #: _____ **Student ID #:** 1022835

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose ***version A****.*

Station #1

Dry flask 10.3253 g NS

Solution flask 17.8287 g NS

Station #2

Alum vial 7.4415 g

Percent yield (%) = Ay / Ty × 100 %

7.4415 / 8.879 = 83.81 %

Station #3

$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$

hydrogen gas made from 0.0475 g Mg

$M_g = 24.31$ (12, 1, 1.2)

$H = 1.008$ (2.1)

2.

Station #4 Calorimetry

Temp 23.00 C.

Station #5

3.6 cm L

1.2 cm w

1.2 cm h

V = L x w x h

Volume

Final V = 5.184 cm

Station #6

Observations: Turned a milky white instantly no other visible reaction.

Reaction Type: Acid/base Neutralization

Balanced Equation: 1mL $CaCl_2$ and Na_2CO_3 ?

Station #7

V 10.00mL NaOH 0.9573M

IB 1.27 mL 17.13 mL

FB 18.40 mL NaOH = 17.13mL

Station #8

Bunsen Burner → To raise the temp of an object or graduated cylinder ↘ use a crucible on it

To accurately measure 25 mL with exact decimal placement

Station #9

25.00 mL

2.5789 M sodium chloride

100.00 mL

Find molarity = ?

Station #10

ME + Lid 21.8563g

MC + Lid + Mg 21.9751g

MC + Lid + Mg Oy compound 22.1715 g 22.1722 g

Oy mass 0.1964g 0.1971g

Extra Space

CHEM 3A – Lab Practical – VERSION A **Name:** Khushpreet Kaur Brar

Section #: Tuesday 6:30 PM **Student ID #:** 10540023

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose ***version A****.*

Station #1

empty flask → 9.5860 g

→ 19.8141 g = $\frac{9.5860\text{g}}{19.8141\text{g}}$

⇒ 0.48389

Station #2

$\frac{7.4415\text{ g}}{8.879\text{ g}} \times 100 = 83.81\%$

Station #3

$Mg(s) + 2\,HCl(aq) \rightarrow MgCl_2 + H_2(g)$

0.0475 g → magnesium

= 0.003944 g

Station #4

23.0 C°

Station #5

3.6 cm × 1.2 cm × 3.6 cm

V = 15.552

Station #6 Na_2CO_3 & $CaCl_2$

Observations: The mixuture of Na_2CO_3 & $CaCl_2$ turned into a smoky & cloudy reaction.

Reaction Type: Precipitation

Balanced Equation: $Na_2Cl_2 + CaCO_2 (aq)$

Station #7

17.5ml — 10.00 ml — volume — 0.9573M

1.27 ml - initial buret

final buret reading. → 1.27 ml

Noah concentration

Noah concentration used

Station #8

⇒ Bunsin burner

⇒ graduated cylinder.

Station #9

final molarity of solution 25.00ml of a 2.5784M sodium chloride

$$\Rightarrow \frac{1.578\,M \times 25.00\,mL}{100.00\,mL} \Rightarrow \frac{39.45}{100.00}$$

⇒ 0.3945M

Station #10

empty 21.8563 g

− 21.9751 g

0.1188

22.1715

− 22.1722 g

− 0.007

mass of oxygen reacted. → 0.1971g

Extra Space

CHEM 3A – Lab Practical – VERSION A **Name:** Leah.A

Section #: _____ **Student ID #:** 1012027

*During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version A**.*

Station #1

19.9023

10.3534g – mass

10.3534
- 19.9023
9.5489g

9.5489 g

Station #2

Percent yield % = (Actual yield)(theoritical yield) x100

Most Alum a student could have made is 8.879 g

mass of vial A – 7.4415g

Calculate percent yield

$\frac{7.4415g}{8.879} \times 100 = 83.810$

83.819

Station #3

Chemical eq – $Mg(s) + 2HCl(aq) \Rightarrow MgCl_2(aq) + H_2(g)$

Calculate mass of hydrogen gas (in grams) that could be made if a student used a 0.0475 piece of Mg

31.7

0.0475

1.505

1.519

Station #4

Add 25mL of water to plastic cup, measure temp

Temp 23°C

Station #5

Determine volume $V = l \times w \times h$

0.9cm

1/2

1 3/4cm

3.5 x 0.9 x 3.5 = 11.025

W

L

V = 11.025

Station #6

Observations: cloudy, White, Residue left $Na_2CO_3 + CaCl_2$

Reaction Type: Precipitation

Balanced Equation: $3CaCO \rightarrow 2NaCl_2$

Station #7

$H_3CC_4H_3O_7(aq) + 3NaOH(aq) \rightarrow Na_3CC_4H_5O_7(aq) + 3H_2O(l)$

NaOH concentration - 0.9573M 10.00mL

Trial 1

Volume 10.00mL
Inital 1.27mL
Final

Calculate volume of NaOH used Record NaOH used

1.18mL

Station #8

Volumetric Flask - records measurments of Liquids in mL

Bunsen Burner - used to connect to gas to burn chemicals

Station #9

Whats the final molarity of a solution made by diluting 25.00mL of a 2.5789M Sodium chloride solution to a final volume of 100.00mL

25.00mL
2.5789 M
100.00mL

25.00 x 2.5789 = 64.4725
9.6940
100

64.5M

969.41mL

Station #10

Mass of oxygen? that reacted

Mass of crucible + Lid empty - 21.8563g

Mass of crucible + Lid + Mg - 21.9751g

Mass of crucible + Lid + Mg x ox compound - 22.1715g
22.1722g

Mg - 24.31
O - 16.00

21.9751g
- 22.1715
0.1964g

0.1964g

Extra Space

CHEM 3A – Lab Practical – VERSION A **Name:** Brianna Sanchez

Section #: Tues **Student ID #:** 1065787

*During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version A**.*

Station #1

mass = 19.9029 g
volume = 10.00 mL
$\frac{19.9029\ g}{10.00\ mL} = 1.99029\ (\frac{m}{v})$

Station #2

8.879
7.4415 g
(Actual yield) / Theoretical yield) × 100
(7.4415)/(8.879) × 100 = 83.84953261
83.84%

Station #3

$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$
24.31 + 70.9 + 2.016 + 0.0475 = 97.2735 g

Station #4

23.5 K

Station #5

height 0.9
lenghth 3.4
width 0.9
$V = l \times w \times h$
$V = 0.9 \times 3.4 \times 0.9 = 2.754\ V$

Station #6

Observations: the top part was separating a bit, fogged up and sizzled

Reaction Type: ?

Balanced Equation: ?

Station #7

Final Buret reading = 18.4 mL

$\frac{18.4\text{mL} - 1.27\text{mL}}{10.00\text{mL}} = 1.713\text{V}$

Station #8

1. Graduated Cylinder
2. Bunsen Burner

Station #9

$100.00 \frac{2.5789\text{M}}{25.00\text{mL}} = 10.3156\text{M}$

Station #10

$22.1722 - 21.9751 = 0.1971\text{g}$

Extra Space

CHEM 3A – Lab Practical – VERSION A **Name:** Mylissa Lopez

Section #: TUE 6:30 **Student ID #:** 06388910

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose ***version A****.*

Station #1

18.1199g

23.5143g

$\frac{\text{mass}}{V}$

$\frac{10.1394\text{g}}{10\text{mL}} = 1.0139\text{g}$

Station #2

8.879 g Alum must have

7.4415g version A

$\frac{7.4415\text{g}}{8.879\text{g}} = 83.8101\%$

Station #3 calculate mass of Hg in grams

0.0475 g

$Mg(s) + 2HCl(g) \rightarrow MgCl_2 + H_2$

24.31 avg mass

1.1547 g

Station #4

22.2 C

~~22.2 + 273.15 = 295.35 K~~

Station #5

L 1.5

W .5 .375

H .5

Station #6

Na_2CO_3 $CaCl_2$

Observations:

Solution turned milky white

Reaction Type:

Precipitation

Balanced Equation:

$NaCl_3$ CO_3Ca

Station #7 Calculate volume of NaOH used

1.27 mL initial 10.00 mL titrant

17.59 mL Final NaOH .9 17.59 − 1.27 = 16.32

Station #8

bunsen burner

volumetric cylinder

Station #9

molarity of solution

25. mL of 2.5789 M Sodium chloride

100. mL final volume. $\frac{2.5789\,M \times 25\,mL}{100\,mL} =$ .6447 mol

Station #10

Crics w/lid 21.8585

Lid tmg 21.9751

Lid tmg ox 22.1715 .1964g

22.1722

Extra Space

CHEM 3A – Lab Practical – VERSION A **Name:** Arely Sequeiros

Section #: 19320 **Student ID #:** 0880073

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose ***version A****.*

Station #1

~~$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$~~

$D = \frac{mass}{volume}$

Dry 10.00mL flask 10.3256g

w/ water 24.9018g

$D = \frac{14.5762g}{10.00mL} = 1.45762 = 1.46g/mL$

$m = 24.9018g - 10.3256g = 14.5762g$

Station #2

percent yield(%) = (actual yield)/(theoretical yield) x 100%.

(7.4415g)/(8.879g) x 100%.

0.8381 x 100%.

percent yield(%) = 83.8%.

Station #3

$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$

calculate mass of H_2 (gram) if 0.0475g of magnesium

12 1

Mg H

24.31 1.008(2)

2.016g H_2

Station #4

22.2°C

Station #5

3.6 x 1.2 x 1.2 = 5.184

$V = 5.18cm^3$

Station #6

Observations: changed to white color

Reaction Type: ~~Redox~~ Acid/Base neutralization

Balanced Equation: $Na_2CO_3 + CaCl_2 \rightarrow Na_2Cl_2 + CO_3Ca$

Station #7

(NaOH) initial: 1.27
final buret reading: 17.55mL

17.55 - 1.27 = 16.28

Volume of NaOH? → 16.28mL

0.9573M

Station #8

Bunsen Burner
Graduated cylinder
↳ measures liquids

Station #9

Molarity of solution made by diluting 25.00mL of 2.5789M sodium chloride to a final volume of 100.00mL?

$$M = \frac{2.5789M}{100.00mL - 25.00mL} = \frac{2.5789M}{75.00mL} \times \frac{1000mL}{1L} = \frac{2578.9M}{75.00} = 34.39M/L$$

Station #10

Mass from balance:

1. empty crucible + lid 21.8563g
2. crucible + lid + Mg 21.9751g
3. crucible + lid + Mg_xO_y 22.1715g 1st
   22.1722g 2nd

22.1722 - 22.1715

mass of oxygen that reacted?

0.0007g

Extra Space

CHEM 3A – Lab Practical - VERSION B **Name:** BRIALYN RULLAN

Section #: 19320 **Student ID #:** 1021678

*During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version B**.*

Station #1

46.9067g (EMPTY)
51.9344g (NACl)
100.1147g (SOLUTION A)

100.1147g
−46.9067g
53.2089

$\frac{53.208}{51.9344g} \times 100 = 102.45232g$

Station #2

24.7857g (MASS OF THE ALUM B)

THIS MEASUREMENT SHOULD BE THEORETICAL MEASURING ASSUMING THEY BOTH WEIGH THE SAME?

Station #3

RECORD THE GAS:
=32.04
36.477 (VAPOR PRESSURE BELOW 100°)

0.082058 L·atom

Station #4

THE TEMPERATURE LOOKED LIKE IT WAS GOING DOWN (GOING DOWN SLOWLY)
ENDOTHERMIC

Station #5

3.5 cm (VERSION B) cm → mm
x100
350

10^{-3} milli
39.37 METER

Station #6

Observations: FIZZED THE FIRST few seconds, THEN DISAPPEARED (HAD VERY LITTLE BUBBLES)
Reaction Type: GAS EVOLUTION

Balanced Equation: CO

Station #7

VOLUME OF "6 DOWN" SODA — 10.00 mL ($H_3C_6H_5O_7$)
INITIAL BURET READING (NaOH) — 1.27
FINAL BURET 17.12 mL
NaOH — 0.9573M

MOLARITY = (MOL SOLUTE)(SOLUTION)

1.5770 = 0.121527

$H_3C_6H_5O_7(aq) + 3NaOH$

Station #8

TEST TUBE, VOLEMETRIC FLASK 125 mL

1) THE TEST TUBE IS USED FOR ADDING SOLUTION, AS WELL AS USING THEM FOR EXPERIMENT
2) THE FLASK IS USED FOR MEASURING SOLUTIONS and SUBSTANCES.

Station #9

GRADUATED CYLINDER — 32.02 mL
TEMPERATURE OF WATER — 22.93 degC

Station #10

mass of Mg used: 0.1524 g
mass of the MgxOy compound: 0.2527 g
mass of oxygen in MgxOy compound 0.1003 g

mass of oxygen =

$\frac{0.1003}{0.1524} = 0.65813$

$\times 0.2527$

0.166371

Extra Space

CHEM 3A – Lab Practical - VERSION B **Name:** Amir Rana

Section #: 19320 **Student ID #:** 1060876

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose ***version B****.*

Station #1 mass vol of solution A mass vol percent $\left(\frac{m}{v}\%\right)$ = mass

(empty) 46.9067g
+NaCl 51.9344g
+solution A 100.1147g

Station #2

mass of empty vial 15.5099g
vial + alum 24.7851g
mass of alum:
24.7851g − 15.5099g = 9.2752g

Actual yield?

Station #3

31.18 mL

×10

mass volume percent $\left(\frac{m}{V}\%\right)$ = $\frac{\text{mass solute g}}{\text{mL solution}}$

Station #4 mass-volume of

go down
exothermic or add ammonium?

Station #5

3.56 mL
convert to mm:

Station #6

Observations: bubbles

Reaction Type: gas evolution

Balanced Equation: $HCl + Na_2CO_3$

Station #7 molarity (M) = (mol solute)/(L solution)

Initial Buret Reading 1.27 mL
Volume "lo down" soda 10.00 mL
Final 17.12 mL

Station #8

- Erlenmeyer flask
- Test tube

Station #9

31.9 mL
22.5 C°

Station #10

mass of Mg 0.1524g
mass " Mg_xO_y 0.2527g
" " Oxygen 0.1003g

$$\frac{0.1003\,g}{0.2527\,g} \times 100$$

= 39.69%

Extra Space

CHEM 3A – Lab Practical - VERSION B **Name:** Ivan C. Garcia

Section #: 19320 **Student ID #:** 1053780

*During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version B**.*

Station #1

50 mL empty 46.9067g 5.0277g

+ NaCl = 51.9344g 53.208g

solution A = 100.1147g 1.2736

9.45%

Station #2

empty = 15.5101g 24.7863 − 15.5101 = 9.2762g

Product = 24.7863 Actual yield

Station #3

35.00 ml 19.827 mm Hg

Station #4

started at 21°C went down to 20°C

reaction was endothermic

Station #5

3.4 cm = .34mm

$3.4\text{cm} \times \frac{1\text{cm}}{10\text{mm}} = .34\text{mm}$

Station #6

Observations: both mixed stayed clear, no reaction

Reaction Type: no reaction

Balanced Equation: $Na_2CO_3 + HCl \rightarrow Na_2Cl + H_3CO$

Station #7

0.9573 m NaOH

1.27 → 17.12 ml

10 ml 6 down 15.85

16.56 ML

Station #8

Beaker

test tube

Station #9

31.00 mL

23.0 °C

Station #10

mg — 0.1524 — g

MgO_y — 0.2527 — g

mass O_2 in MgO_y — 0.1003 — g

what % O_2 in MgO_y

39.69%

Extra Space

CHEM 3A – Lab Practical - VERSION B **Name:** Cristina A. Perez-Rodriguez

Section #: 19320 **Student ID #:** 1090762

*During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version B**.*

Station #1

→ mass - volume % = 9.45 %

Station #2

- empty container: 15.5098 g
- Container + Alum: 24.7851 g

→ mass of Alum: 9.2753 g

→ this would be the Actual Yield

$$\begin{array}{r} 24.7851 \\ -\ 15.5098 \\ \hline 9.2753 \end{array}$$

Station #3

→ volume of gas collected: 31.51 mL

Station #4

→ The temp decreased making the reaction exothermic.

Station #5

→ The cylinder measured 3.5 cm at its longest & when converted to mm it measured 35 mm.

Station #6 Na_2CO_3 + HCl ← Strong acid

Observations: No temperature change, no precipitation, bubbles at first contact but did not last

Reaction Type: Acid / Base Neutrulization

Balanced Equation: $Na_2CO_3(aq) + HCl(aq) \rightarrow Na_2Cl(aq) + HCO_3(aq)$

Station #7

→ Molarity = 60.40 M

Station #8

→ Elenmyer Flask

→ test tube

Station #9

→ Graduated Cylinder : 32.1 mL

→ Temperature : 22.8°C

Station #10

→ percent of oxygen : 39.69%

Extra Space

CHEM 3A – Lab Practical - VERSION B **Name:** Elizabeth Gonzalez

Section #: TUE 19320 **Student ID #:** 016037103

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose ***version B****.*

Station #1

(empty) stopper + vial 46.9067 g

stopper + solution "A" 51.93744 g

100.1147 g

$\left(\frac{m}{v}\%\right) = \frac{\text{mass solute}}{\text{mL solution}} \times 100$

$\frac{100.1147\text{ g}}{51.93744\text{ g}} \times 100 = 192.7714578\text{ mL}$

Station #2

version B 24.7860 g (determine mass) actual yield, theoritical or percent yield?

%. = x100%.

$247.7860 - 15.5114 = 9.2746 \times 100 = 927.46$

empty 15.5114 g

Alum = 474.46 g/mol

Station #3

744.4 mmhg Record gas

Station #4

there was no reaction on thermometer, stayed the same

Station #5

3.5 cm convert to mm

$3.5\text{ cm} \times 10 = 35\text{ mm}$

Station #6

Observations: Nothing happened, solids remained the same.

Reaction Type: NO reaction

Balanced Equation: $Na_2CO_3(aq) + HCl(aq)$

complete with phase symbols

Station #7

(molarity) = (mol solute) / (L solution) 10.00 mL "6 down soda"

0.9573 M = 1.27 mL

= 17.12 mL final is ($H_3C_6H_5O_7$)

$H_3C_6H_5O_7(aq) + 3\,NaOH(aq) \rightarrow Na_3C_6H_5O_7(aq) + 3H_2O(l)$

Station #8

~~test~~ tube

erlenmyer flask

Station #9

32 mL

22.5 C°

Station #10

mass of Mg used 0.1524 0.2 = 0.1003 g

mass of MgxOy compound 0.2527 = 39.69% g

mass of oxygen in MgxOy compound 0.1003 g

Extra Space

0.2527 - 0.1524 = 0.1003 g

0.1003 ÷ 0.2527 x 100 39.69

CHEM 3A – Lab Practical - VERSION B **Name:** Evelyn Garcia

Section #: TUESDAY 6:30 pm **Student ID #:** 0951887

*During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version B**.*

Station #1

Mass-volume % of Solution A = ? Solution NaCl in a 50.00mL flask

mass of 50 mL v flask + stopper empty = 46.9067 g
mass of 50 mL v flask + stopper + NaCl = 51.9344 g
Mass of 50 mL v flask + stopper + solution A = 100.1147 g

$51.9344 - 46.9067 = \frac{5.0277}{50.00} \cdot 100 = 10.0554\ \%$

Station #2

Mass of Alum = 24.7854 g
same as mass of empty vial = 24.7854 g
This measurement is actual yield

Station #3

$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$
Volume of gas collected =

Station #4

After adding tablespoon the measurement decreased making it exothermic

Station #5

Oval object 3.5 cm -> ? mm (millimeters)

3.5 cm × 10 = 35 mm

Station #6

HCl and Na_2CO_3

Observations: Immediate bubbling

Reaction Type: Acid base reaction

Balanced Equation: $Na_2CO_3(aq) + 2HCl(aq) \rightarrow 2NaCl(aq) + 1$

Station #7

$H_3C_6H_5O_7(aq) + 3NaOH(aq) \rightarrow Na_3C_6H_5O_7(aq) + 3H_2O(l)$

M = 0.9573

Initial Volume: 1.27 mL

"6" Down soda: 10.00 mL

Final V = 17.12 mL

?? molarity of citric acid $H_3C_6H_5O_7$ in "6 down" soda

Station #8

Erlenmeyer Flask

- Use to measure/mix chemicals

Test Tube

- Use to mix/test chemicals

Station #9

Volume of water in graduated cylinder: 32. mL

Temperature of water in flask: 22.8 C°

Station #10

Mass Mg used: 0.1524 g

Mass Mg_xO_y compound: 0.2527 g

Mass Oxygen in Mg_xO_y compound: 0.1003 g

mass % of oxygen in Mg_xO_y?

$\frac{0.1003}{0.2527} \times 100$

0.396913 · 100

39.69 %

Extra Space

CHEM 3A – Lab Practical - VERSION B **Name:** Yordaamir Kaur

Section #: T6:30PM **Student ID #:** 1054041

*During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version B**.*

Station #1

Mass - volume % of sol'A

100.1147g − 46.9067g = 53.208

= $\frac{53.208}{50.00} \times 100$ = 106.416 g/ml

Station #2

Mass of Alum is 9.2748g

It theoretical yield.

Station #3

Mass of H_2 = 2.016

V = m x d

V = 2.016 x

Station #4

1. Temperature went down.
2. Endothermic Reaction

Station #5

1. 3.45 cm
2. 34.5 mm

Station #6

Observations: After mixing Na_2CO_3 and HCl, bubbles were formed + becomes cloudy

Reaction Type: Gas evolution, Double displacement Rxn

Balanced Equation: $Na_2CO_3 + HCl \rightarrow NaCl_2 + H_2O + CO_2$

Station #7

Molarity = $\frac{mol}{L}$

initial Buret = 1.27 ml

Final Buret = 17.12 ml

Volume = 10.00 ml = 0.0100 L

Station #8

Erlenmeyers Flask and Test Tube

Station #9

volume of water - 31.9 ml

Temperature - 22.4°C

Station #10

mass of compound = 0.15424 g

mass of Oxygen = 0.1003 g

Mass % = $\frac{0.1003\,g}{0.15424\,g} \times 100 = 39.6913\%$

Extra Space

CHEM 3A – Lab Practical - VERSION B **Name:** Alexander Alvarado

Section #: 19320 **Student ID #:** 1043122

*During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version B**.*

Station #1

51.87% = mass volume %

$\frac{5.2777}{53.208} \times 100 =$

Station #2

15.5103 g - empty
24.7864 g - full w/ bottle
= 9.2761 g mass of alum

Actual yeild

Station #3

The volume of gas collected is 17.753 mmhg

Station #4

The temp went up and it was exothermic

Station #5

3.4 cm object b

convert to mm?

Station #6

Observations: No reaction happened

Reaction Type: No reaction

Balanced Equation: Na^2HClCO^3

Station #7

10.00 mL = 1.27 mL / 17.12 mL

M = 0.9573 M

Station #8

Volumeric Flask and Test tube

Station #9

Volume of graduated cylinder: 31.0 mL
Tempature of water in the flask: 20.2 C°

Station #10

5.03% Mg_xO_y

Extra Space

CHEM 3A – Lab Practical - VERSION C **Name:** Xinaya Ocampo

Section #: _____ **Student ID #:** 09728674

*During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version C**.*

Station #1

Molarity = (mol solute)/(L solution) Molarity = 10.046

58.44277 g/mol 5.0234g of NaCl

50mL "solution A"

Station #2

Mass of Aluminum foil = 0.3801 g

26.98 g/mol / 0.3801 g = 10.2551 g

= 10.2550 98

Station #3

21.10 °C = Temp of Water

745.1 mmHg = P total

745.1 = P_{H_2}

Station #4

$q = m \cdot C \cdot \Delta T$

35.0392 g x 4.184 J/g°C x 21.1°C = 3093.34467

q = 3093.34

Station #5

3.8 x 1 x 1.9 = 7.22 cm

mL → 21.66

Station #6

Observations: Turned milky white instantly

Reaction Type: Acid/Base Neutralization

Balanced Equation: $BaCl_2 + Na_3PO_4$

Station #7

18.45 mL

Station #8

- Looking glass
- Used for measuring a liquid

Station #9

3.85 cm

Station #10

Mass of Mg 0.1495 g (Mass int)

Mass of MgxOxy Compound: 0.2479 g

0.0984 g ← Mass of Oxygen in MgxOxy compound

24.79% + 9.84% = 34.63%

0.3463

Extra Space

CHEM 3A – Lab Practical - VERSION C **Name:** Noah cruz

Section #: 15 – 3A **Student ID #:** 10444628

*During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version C**.*

Station #1

M = (mol solute)/(L solution)

5.0234 g NaCl

molecular mass of NaCl = 58.44277 g/mol

50.00mL × 1L/1000mL = 0.05L

5.0234g × 1 mol / 58.44277 g/mol = 0.08595 g/mol

M = 1.719

Station #2

mass of Aluminum: 0.3807g

molar mass of Al: 26.98 g/mol

molecular mass of Alum: 474.44 g/mol

maximum amount of Alum in g = 0.000029749 or 0.1174g

0.3807g Al × 1/26.98 g/mol × 1 Alum / 1 Aluminum × 1/474.44 g/mol Alum

Station #3

Record Temp of water: 20.0°C

Record room pressure (total) = 744.8 mmHg

1 mmHg = 1 Torr

1.3352 × 10² Pa

$P_{total} = P_{H_2} + P_{H_2O}$ (in H_2)

=

in the collecting tube

Station #4

mass NaOH added	36.25g
Total solution(m)	35.03 g (92)
Specific heat (c)	4.184 J/g°C
Change in temp	21.1°C

q = m x C x ΔT

q = 3092.53 J/g°C

Station #5

V = of C by water displacement

initial = 21.0mL final = 28.1

28.1 − 21.0 = 7.1mL

Volume of object C = 7.1ml

Station #6

Observations: Turns white instantly, solids form $Ba^{+}Cl_2^{-} + Na_3^{+}PO_4^{-}$

Reaction Type: Double replacement

Balanced Equation: $3\,BaCl_2 + 1\,Na_3PO_4 \rightarrow 1\,Ba_3PO_4 + 3\,NaCl_2$

Station #7

final buret reading = 18.4mL

Station #8

C = Volumetric buret/pipet

C = medium center glass

Station #9

line on paper in cm = 3.8 cm

Station #10

% of mg in MgXOy

mass of mg : $\frac{0.1495\,g}{0.2478\,g} \times 100 = 60.31\%$ mg in MgXOy

mass of MgXOy :

Extra Space

CHEM 3A – Lab Practical - VERSION C **Name:** Steffany Sinsuan

Section #: _____ **Student ID #:** 0073521

*During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version C**.*

Station #1

weigh 5.0234g of NaCl
molar mass of NaCl is 58.44277 g/mol
$\frac{58.44277 \times 5.0234}{50.00} = 5.87011\ M$

Station #2

0.3806g
maximum amount of alum? in grams
Alumfoil 26.98 g/mol
AWm 474.46 g/mol } molar mass

Station #3

21.0 = temp of water
ptotal = 24.2 $\frac{21.0}{24.2} = 0.86$
$p_{total} = P_{H_2} + P_{H_2O}$

Station #4

$q = m \times C \times \Delta T$
mass of NaOH added 30.25g
mass total solution (m) 35.0392 g
Specific heat s (c) 4.184 J/g°C
chang in temp (ΔT) 21.1°C

$35.0392g \times 4.184 \times 21.1$
$q = 3093.34467$

Station #5

23.15 mL

Station #6

Observations: When mixed together, it first made the solution cloudy but the solution soon became clear.

Reaction Type: No Reaction Na_3PO_4 $BaCl_2$

Balanced Equation: $Na_3PO_4(aq) + BaCl_4(aq) \rightarrow 3NaCl_4(aq) + 2BaPO_4$

Station #7

17.58 mL

Station #8

Volumetric pipet, Glass watch

Station #9

1.50 in / 3.88 cm

Station #10

Mass % of magnesium: 9.84 %. $0.0484 - 0.1495 = 0.0989$

Mass of Mg 0.1495g

Oxygen in Mg_xO_y 0.0989

$\frac{\times 100}{9.84}$

Mg_xO_y 0.2479

Extra Space

d $M_1V_1 = M_2V_2$ $D = \frac{m}{V}$

CHEM 3A – Lab Practical - VERSION C M Moles / Liters **Name:** Tayler Rios

Section #: _____ **Student ID #:** 0934356

*During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version C**.* $P_1V_1 = P_2V_2$

$V = V_F - V_i$

Station #1 $q = m \cdot c \cdot \Delta T$

50.00mL Solution a 5.0234g NaCl

$\frac{58.44277 g/mol}{5.0234 g}$ 11.634M

mass % = _____ x100

Station #2

M of alum

0.3806g $\frac{0.3806g}{26.98 g/mol}$ = 0.01411 mol

max amt of alum(g) can be made

molar mass

Alum F (26.98g/mol)

Alum (474.46g/mol)

Station #3

22.1°C change 2 kelvin + 273.15 = 295.25K

745.1 atm Pressure

P of hydrogen 32.31mL

$P_{total} = P_{H_2} + P_{H_2O}$

327.56mm Hg = 32.31mL + 295.25K

Station #4

30.25g

35.0392g · 4.184 J/g°C · 21.1°C = 3093.3J

↑

(or) $q = m \cdot c \cdot \Delta T$

Station #5 heat abs

~~250~~mL water

37.1mL

40.1mL added = 3.1mL

37.1mL − 40.1mL = 3.1mL

Station #6

Observations: once mixed products became milky white

Reaction Type: redox

Balanced Equation: $BaCl2 + Na3PO4 \rightarrow BaCl_2Na_3PO_4$

Station #7

17.58mL

Station #8

watch glass

volumetric pipet

Station #9

3.88 cm

Station #10

mass Mg 0.1495g

mass MgOy 0.2479g } sub

mass Oxygen 0.09841g

$\frac{0.1495}{0.2479} \times 100 = 58.85g$

Extra Space

~~FF~~

$\frac{(0.1495g)}{(0.2479g)} \times 100$

$\frac{1.}{0.1495}$

CHEM 3A – Lab Practical - VERSION C **Name:** Shantal

Section #: _____ **Student ID #:** 1047747

*During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version C**.*

Station #1

make 50.mL Sol A
5.0234g of NaCl
molar mass of NaCl = 58.4427 g/mol

58.44277g/mol - 50.00 = = 8.44277 g/mol

Station #2

0.3805g of Al

= 0.3805g is ~~the total amount of Alum.~~
that can be made

Station #3

Temp of water: 20.1 °C
Record pressure in classroom: 744.5
Pressure Hydrogen 37.00

$P_{total} = P_{H_2} + P_{H_2O}$

17.535 mmHg

Station #4

mass of NaOH 30.25g
mass total Solution(m) 35.0392g
Specific Heat of Solution(c) = 4.184 J/g°C
change in temp: ΔT = 21.1°C

$q = m \times C \times \Delta T$ $35.0392g \times 4.184 \frac{J}{g°C} \times 294.1K$

= 43116.24

Station #5

21.00 mL

$BaCl_2$, Na_3PO_4

Station #6

Observations: Cloudy

Reaction Type: Acid/base Neutralization

Balanced Equation: $BaCl_2 + Na_3PO_4 \rightarrow Na_3Cl_2 + PO_4Ba$

Station #7

record the final buret reading

= 18.40 mL

Station #8

c) glass flask

c) pipet use to measure 10.00 mL to certain flask or beaker.

Station #9

3.08cm

$P = \frac{\text{mass solute}}{\text{mL solut}} \times 100$

Station #10

Mass of Mg used: 0.1495g $\left(\frac{Mg}{24.31}\right)\left(\frac{O}{15.999}\right)$

Mass of MgxOy: 0.2479 g

Mass of oxygen in MgxOy compound: 0.0984g

% percent of mag

$\frac{0.1495g}{0.2479} \times 100$

= 60.31%

Extra Space

e) glass flask

d) pipet used to give 10.00mL of product to beaker or

CHEM 3A – Lab Practical - VERSION C **Name:** Alexis Romero

Section #: Tues 6:30 **Student ID #:** 0692985

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose ***version C****.*

Station #1 Solutions / Dilutions

Solution "A" 50.00mL

Molarity: $5.0234\,g = \frac{g/mol}{1\,g} \times \frac{1\,mol}{1\,mol} \times \frac{50\,g/mL}{1\,mL} =$ g

Station #2 Synthesis of Alum

Molar Mass

26.98 g/mol - Al

474.46g/mol Alum

0.3801g Mass of Al foil

$\frac{0.3801}{474.46g} =$

Station #3 Gas Constant

20.9°C of H_2O

744.2 mmhg

32.9ml - Buret tube

what is the Pressure of Hydrogen gas in collection tube?

$P_{total} = P_{H_2} + P_{H_2O}$ in mm Hg

Station #4 Calorimetry

$q = m \times C \times \Delta T$

mass of NaOH added 30.25g

Mass Total solution (m) 35.0392g

Specific Heat of Solution (C) 4.184 J/g°C

Change in Temp (ΔT) 21.1°C

Heat (q): 3366.3 kJ

$q = 35.0392g \times 4.184\,J/g \times 21.1°C + 273K$

$q = 3093.34467$

$q = 3366.34467$

Station #5 Making Measurments.

Volume Block C: 3.8cm length x 1cm width x 2cm height

V = 7.6 g/mL

30 mL Initial starting

38 mL Final Difference: is 8mL

Station #6 Types of Reactions

Observations: White coagulation, precipitate $BaCl_2 + Na_3PO_4$ white chunks

Reaction Type: Precipitation - aqueous

Balanced Equation: $Ba_2Na(aq) + Cl_2PO_2(aq)$

Station #7 Determination of Citric Acid in Soda

Final buret 17.48mL

Initial 1.27mL
Final 17.48mL

Vol of 6 Days Soda 10.0 mL

Station #8 General Lab Knowledge

- glass dish/glass plate to place specimen or protect it if its
- Titration tube/pipette for small amounts of liquids only.

Station #9 General Lab Knowledge

3.86cm is the length of the line shown on the paper.

Station #10 Empirical Formula

1. Mass of Mg used — 0.1495 g
2. Mass of MgOxy: 0.2479 g
3. Mass of Ox in MgOxy = 0.0984 g

Mass percent of Mg in MgOxy

$\frac{0.1495}{0.2479} \times 100 = 60\%$

60.3%

Extra Space

CHEM 3A – Lab Practical - VERSION C **Name:** Maritza Madrigal

Section #: _____ **Student ID #:** 10587277

*During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version C**.*

Station #1

5.0234 g
58.44277 g/mol
= 0.08596 mol

Station #2

0.33069 mol Al = 0.3306 g / 26.98 g/mol = 0.01411 mol Al

0.01411 mol × 474.46 g/mol = 6.69 g alumn

Station #3

18.7
744.3 P

P_{H_2} = 744.3 − 18.7
P_{H_2} = 726.1 mm Hg

726.1 mm Hg

Station #4

m = 35.0392 g
c = 4.184 J/g°C
ΔT = 21.1°

35.0392 × 4.184 = 146.404
146.404 × 21.1 = 3089.1
q = 3.09 × 10³ J
(3.09 kJ)

Station #5

21.03 mL
41.00

41.00 − 21.03
= 19.97 mL

Station #6

Observations: it was gray cloudy color.

Reaction Type: Acid/Base Neutralization

Balanced Equation: $BaCl\ Na_3PO_4$

Station #7

17.12 mL

Station #8

Pipet dropper, petri dish

Station #9

3.9 cm

Station #10

$\text{mass \% mg} = \frac{1.1495}{0.2479} \times 100$ ≈ 60.3% mg

$\text{mass \% mg} = \frac{0.1495}{0.2479} \times 100$

$\frac{0.0984}{0.2479} \times 100$

$\frac{0.0984}{0.2479} \times 100$ ≈ 39.7%O

Extra Space